

Nonsquare matrices as transformations between dimensions

Moving between 2D, 3D and beyond.

THE IDEA

A matrix does not have to be square. A 3×2 matrix takes a 2D vector and produces a 3D vector; a 2×3 matrix does the reverse. The method never changes — each column still says where one basis vector lands.

Reading the shape: columns = input dimension, rows = output dimension. A 3×2 matrix has 2 columns (2D input) and 3 rows (3D output).

Shape	Meaning
2×2	2D $\rightarrow$ 2D, stays in the plane
3×2	2D $\rightarrow$ 3D, embeds a plane into space
2×3	3D $\rightarrow$ 2D, projects space down onto a plane
1×2	2D $\rightarrow$ a single number — this is exactly the dot product

WORKED EXAMPLE

CHAPTER 8 — WORKED EXAMPLES

matrices that move between dimensions

HOW TO READ THE SHAPE

rows = output dimension columns = input dimension

a 3×2 matrix : 2 columns $\rightarrow$ input is 2D, 3 rows $\rightarrow$ output is 3D

EXAMPLE — 2D input, 3D output

$$A = \begin{bmatrix} 2 & 0 \\ 1 & 3 \\ 0 & -1 \end{bmatrix} \quad \hat{i} \rightarrow (2, 1, 0) \quad \hat{j} \rightarrow (0, 3, -1)$$

apply to $v = (3, 2)$:

$$3 \times (2, 1, 0) = (6, 3, 0)$$

$$2 \times (0, 3, -1) = (0, 6, -2)$$

$$\text{add : } (6+0, 3+6, 0+(-2)) = (6, 9, -2)$$

a 2D vector became a 3D vector

SHAPES AT A GLANCE

2×2 2D $\rightarrow$ 2D (same space)

3×2 2D $\rightarrow$ 3D (into higher dim)

2×3 3D $\rightarrow$ 2D (projection down)

1×2 2D $\rightarrow$ 1 number (this is the dot product)

Why it matters for ML: every neural network layer is a nonsquare matrix. A layer mapping 768 features to 256 is a 256×768 matrix — the same operation you just did by hand, at scale.

TEST — Chapter 8

1. A 3×2 matrix with $\hat{i} \rightarrow (1, 0, 2)$, $\hat{j} \rightarrow (0, 1, -1)$. Apply it to $(2, 3)$. $\rightarrow (2, 3, 1)$
2. A 2×3 matrix with $\hat{i} \rightarrow (1, 1)$, $\hat{j} \rightarrow (0, 2)$, $\hat{k} \rightarrow (-1, 0)$. Apply it to $(1, 2, 3)$. $\rightarrow (-2, 5)$
3. A layer maps 5 inputs to 3 outputs. What is the matrix shape, and how many numbers does it hold? $\rightarrow 3 \times 5, 15$ numbers

FORMULAS FROM THIS CHAPTER

Shape	rows = output dim · columns = input dim
3×2	2D → 3D
2×3	3D → 2D
1×2	2D → one number (dot product)